Basic Understanding First

Before solving the problem, let's understand **key concepts** and what this question is asking.

1. **What is a Matrix?**

- A **matrix** is just a table of numbers arranged in rows and columns. For example:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

  - The numbers inside the matrix are called **entries**.
  - Here, A is a 2×3 matrix because it has 2 rows and 3 columns.

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2. **What Does $Ax = 0$ Mean?**

- The equation $Ax = 0$ means multiplying the matrix A by a column of numbers x to get the zero vector (a column of zeros):

$$Ax = 0 \implies \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

  - Here, x_1, x_2, x_3 are unknown numbers we want to solve for.
  - Our goal is to find all possible x_1, x_2, x_3 that satisfy this equation.

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3. **What is the Kernel (N_A)?**

- The **kernel** is the set of all solutions to $Ax = 0$.
- If you imagine x as a vector, the kernel tells us which directions x can point in for Ax to become zero.

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4. What is Nullity ($\nu(A)$)?

- The **nullity** is the **number** of independent solutions to $Ax = 0$. It tells us how many "degrees of freedom" there are in the kernel.
- For example:
  - If the nullity is 1, there is only one basic solution (like one direction for x).
  - If the nullity is 2, there are two independent basic solutions.

Let's Solve Part (a) (i)

Matrix Given:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

Step 1: Write the Equation $Ax = 0$

Let:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

The equation becomes:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Step 2: Write Out the Equations

Multiply the matrix A by x:

1. First row:

$$1 \cdot x_1 + 2 \cdot x_2 - 1 \cdot x_3 = 0,$$

   which simplifies to:

$$x_1 + 2x_2 - x_3 = 0. \tag{1}$$

2. Second row:

$$2 \cdot x_1 - 1 \cdot x_2 + 3 \cdot x_3 = 0,$$

   which simplifies to:

$$2x_1 - x_2 + 3x_3 = 0. \tag{2}$$

Step 3: Solve the Equations

To solve, we use **row reduction** (a method to simplify equations step by step):

1. Write the equations as an augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & -1 & 3 & 0 \end{bmatrix}.$$

2. Subtract $2 \times (\text{Row 1})$ from Row 2:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -5 & 5 & 0 \end{bmatrix}.$$

3. Divide Row 2 by -5 to simplify:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

4. Subtract $2 \times (\text{Row 2})$ from Row 1:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

Step 4: Write the Solution

From the simplified matrix:

1. First row:

$$x_1 + x_3 = 0 \implies x_1 = -x_3.$$

2. Second row:

$$x_2 - x_3 = 0 \implies x_2 = x_3.$$

The solution is:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

Step 5: Write the Kernel

The kernel is:

$$N_A = \text{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Step 6: Find the Nullity

- The number of columns $n = 3$.
- The rank (number of non-zero rows in the reduced matrix) $r = 2$.
- Nullity:

$$v(A) = n - r = 3 - 2 = 1.$$

Part (b) (ii)

Now we'll solve it similarly for the matrix:

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ -6 & 3 & -9 \end{bmatrix}.$$

Step 1: Write the Equation $Ax = 0$

Let:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

The equation becomes:

$$\begin{bmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ -6 & 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Step 2: Row Reduce the Matrix

Using similar steps as above:

1. Write the augmented matrix:

$$\begin{bmatrix} 2 & -1 & 3 & 0 \\ 4 & -2 & 6 & 0 \\ -6 & 3 & -9 & 0 \end{bmatrix}.$$

2. Divide Row 1 by 2:

$$\begin{bmatrix} 1 & -0.5 & 1.5 & 0 \\ 4 & -2 & 6 & 0 \\ -6 & 3 & -9 & 0 \end{bmatrix}.$$

3. Subtract $4 \times (\text{Row 1})$ from Row 2 and $-6 \times (\text{Row 1})$ from Row 3:

$$\begin{bmatrix} 1 & -0.5 & 1.5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Step 3: Write the Solution

1. From the matrix:

$$x_1 - 0.5x_2 + 1.5x_3 = 0.$$

Solve for x_1:

$$x_1 = 0.5x_2 - 1.5x_3.$$

2. Write the solution as:

$$x = x_2 \begin{bmatrix} 0.5 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1.5 \\ 0 \\ 1 \end{bmatrix}.$$

Step 4: Write the Kernel

The kernel is:

$$N_A = \text{Span}\left\{ \begin{bmatrix} 0.5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1.5 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Step 5: Find the Nullity

- $n = 3$,
- $r = 1$,
- Nullity:

$$\nu(A) = n - r = 3 - 1 = 2.$$

Final Answer:

1. **(a) (i)**:
    - Kernel:

$$N_A = \text{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

    - Nullity: $\nu(A) = 1$.

2. **(b) (ii)**:
   - Kernel:
$$N_A = \text{Span}\left\{\begin{bmatrix}0.5\\1\\0\end{bmatrix}, \begin{bmatrix}-1.5\\0\\1\end{bmatrix}\right\}.$$
   - Nullity: $\nu(A) = 2$.